Q1 (2)

P (odd no. twice)= P(even no. thrice)

$\Rightarrow {}^{n}C_{2}\left(\frac{1}{2}\right)^{n}={}^{n}C_{3}\left(\frac{1}{2}\right)^{n} \Rightarrow n=5$

Success is getting an odd number then P (odd successes) $=\mathrm{P}(1)+\mathrm{P}(3)+\mathrm{P}(5)$

$={}^{5}C_{1}\left(\frac{1}{2}\right)^{5}+{}^{5}C_{3}\left(\frac{1}{2}\right)^{5}+{}^{5}C_{5}\left(\frac{1}{2}\right)^{5}$

$=\frac{16}{2^{5}}=\frac{1}{2}$

Q2 (6)

Let x, y, z be probability of B_1, B_2, B_3 respectively

$\Rightarrow x(1-y)(1-z)=\alpha$

$\Rightarrow \mathrm{y}(1-\mathrm{x})(1-\mathrm{z})=\beta$

$\Rightarrow z(1-x)(1-y)=\gamma$

$\Rightarrow (1-x)(1-y)(1-z)=p$

$(\alpha-2\beta)p=\alpha\beta$

$(x(1-y)(1-z)-2y(1-x)(1-z))(1-x)(1-y)(1-z)=xy(1-x)(1-y)(1-z)$

$x-xy-2y+2xy=xy$

$x=2y$ ... (1)

Similarly $(\beta-3r)\mathrm{p}=2\beta r$

$\Rightarrow y=3z$ ...(2)

From (1) and (2)

$x=6z$

Now

$\frac{x}{z}=6$

Q3 (2)

Required probability

$=\frac{{}^{5}C_{2}\times 3^{3}}{4^{5}}$

$=\frac{10\times 27}{2^{10}}=\frac{135}{2^{9}}$

Q4 (1)

Probability of not getting intercepted $= \frac{2}{3}$

Probability of missile hitting target $= \frac{3}{4}$

$\therefore$ Probability that all 3 hit the target $= \left(\frac{2}{3} \times \frac{3}{4}\right)^3 = \frac{1}{8}$

Q5 (4)

Based on Baye's theorem

$$\text{Probability} = \frac{\left(160 \times \frac{35}{100}\right)}{\left(160 \times \frac{35}{100}\right) + \left(100 \times \frac{20}{100}\right) + \left(140 \times \frac{10}{100}\right)}$$

$= \frac{5600}{9000}$

$= \frac{28}{45}$

Q6 (3)

Total cases $(4 \times 9 \times 9 \times 9) - (3 \times 9 \times 9)$

Probability $= \frac{(3 \times 9 \times 9) - (2 \times 9) + (8 \times 9 \times 9)}{(4 \times 9^3) - (3 \times 9^2)}$

$= \frac{97}{217}$

Q7 (2)

$p(x = 9) = p(x = 7)$

${}^{n}C_9 \left(\frac{1}{2}\right)^{n-9} \times \left(\frac{1}{2}\right)^9 = {}^{n}C_7 \left(\frac{1}{2}\right)^{n-7} \times \left(\frac{1}{2}\right)^7$

${}^{n}C_9 \times \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^2 \times {}^{n}C_7$

$x + y = n \quad \Rightarrow n = 16$

$p(x = 2) = {}^{16}C_2 \times \left(\frac{1}{2}\right)^{14} \times \left(\frac{1}{2}\right)^2$

$= {}^{16}C_2 \times \left(\frac{1}{2}\right)^{16} = \frac{15}{2^{13}}$

Q8 (3)

$n(s) = \frac{7!}{2!3!2!}$

$n(E) = \frac{6!}{2!2!2!}$

$$P(E)=\frac{n(E)}{n(S)}=\frac{6!}{7!}\times\frac{2!3!2!}{2!2!2!}$$

$$\frac{1}{7}\times 3=\frac{3}{7}$$

Q1: 24 Feb (Shift 1) - Single Correct

An ordinary dice is rolled for a certain number of times. If the probability of getting an odd number 2 times is equal to the probability of getting an even number 3 times, then the probability of getting an odd number for odd number of times is :

(1) $\frac{3}{16}$

(2) $\frac{1}{2}$

(3) $\frac{5}{16}$

(4) $\frac{1}{32}$

Q2: 24 Feb (Shift 1) - Numerical

Let $B_i (i=1,2,3)$ be three independent events in a sample space. The probability that only B_1 occur is α, only B_2 occurs is β and only B_3 occurs is γ. Let p be the probability that none of the events B_i occurs and these 4 probabilities satisfy the equations $(\alpha-2\beta)p=\alpha\beta$ and $(\beta-3\gamma)p=2\beta\gamma$ (All the probabilities are assumed to lie in the interval (0,1)). Then $\frac{P(B_1)}{P(B_3)}$ is equal to

Q3: 24 Feb (Shift 2) - Single Correct

The probability that two randomly selected subsets of the set \{1,2,3,4,5\} have exactly two elements in their intersection, is:

(1) $\frac{65}{2^7}$

(2) $\frac{135}{2^9}$

(3) $\frac{65}{2^8}$

(4) $\frac{35}{2^7}$

Q4: 25 Feb (Shift 1) - Single Correct

When a missile is fired from a ship, the probability that it is intercepted is $\frac{1}{3}$ and the probability that the missile hits the target, given that it is not intercepted, is $\frac{3}{4}$. If three missiles are fired independently from the ship, then the probability that all three hit the target, is:

(1) $\frac{1}{8}$

(2) $\frac{1}{27}$

(3) $\frac{3}{4}$

(4) $\frac{3}{8}$

Q5: 25 Feb (Shift 2) - Single Correct

In a group of 400 people, 160 are smokers and non-vegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are $35\%, 20\%$ and 10% respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is:

(1) $\frac{7}{45}$

(2) $\frac{8}{45}$

(3) $\frac{14}{45}$

(4) $\frac{28}{45}$

Q6: 25 Feb (Shift 2) - Single Correct

Let A be a set of all 4 -digit natural numbers whose exactly one digit is 7 . Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is:

(1) $\frac{1}{5}$

(2) $\frac{2}{9}$

(3) $\frac{97}{297}$

(4) $\frac{122}{297}$

Q7: 26 Feb (Shift 1) - Single Correct

A fair coin is tossed a fixed number of times. If the probability of getting 7 heads is equal to probability of getting 9 heads, then the probability of getting 2 heads is :

(1) $\frac{15}{2^{12}}$

(2) $\frac{15}{2^{13}}$

(3) $\frac{15}{2^{14}}$

(4) $\frac{15}{2^{8}}$

Q8: 26 Feb (Shift 2) - Single Correct

A seven digit number is formed using digit 3,3,4,4,4,5,5 . The probability, that number so formed is divisible by 2, is :

(1) $\frac{6}{7}$

(2) $\frac{4}{7}$

(3) $\frac{3}{7}$

(4) $\frac{1}{7}$

Answer Key

Q1 (2)	**Q2** (6)	**Q3** (2)	**Q4** (1)
Q5 (4)	**Q6** (3)	**Q7** (2)	**Q8** (3)